Speed Measurement

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Objective

understanding Doppler concept and speed measurements in radar and EW

Learning Outcomes

At the end of the lesson, you should be able to

1. describe Doppler effect and its use in speed measurement
2. describe radar operation employing Doppler phenomena in measuring speed
3. describe spectral behavior of Doppler radars for air platform
4. compare PRI (Pulse Repetition Period) types, and understand how they are utilized in EW systems

Presentation Plan

- Doppler effect
- MTI and Pulse Doppler (PD) radar
- PRI Types
- Spectral Clutter for Airborne pulsed radar

What happens when the target is in motion?

- Clutter echoes are much stronger than typical aircraft echoes !
- Design a filter system, based on doppler effect, at the radar in order to
  - pass echo signals from a target in motion (extract velocity info if possible?)
  - attenuate echo signals from fixed or slowly moving targets (clutter)
- How?

Total *phase change in* $2R$*:* $\varphi = 2\pi\frac{2R}{\lambda} = \frac{4\pi R}{\lambda}$

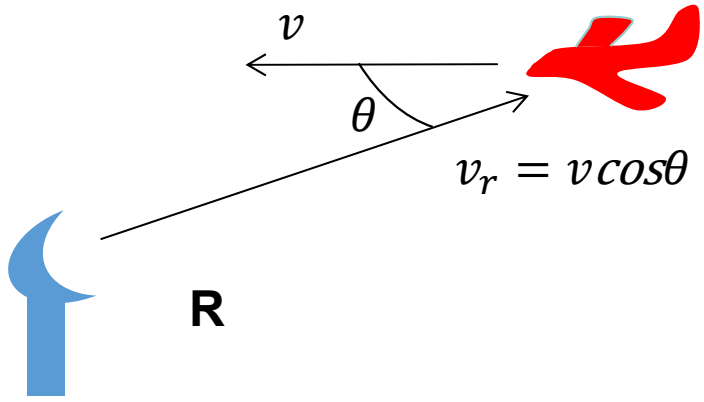
Moving target $\Rightarrow$ *changing* $R \Rightarrow$ *changing phase*

$$\omega_d = \frac{d\varphi}{dt} = \frac{4\pi}{\lambda}\frac{dR}{dt} = \frac{4\pi}{\lambda}v_r = 2\pi f_d$$

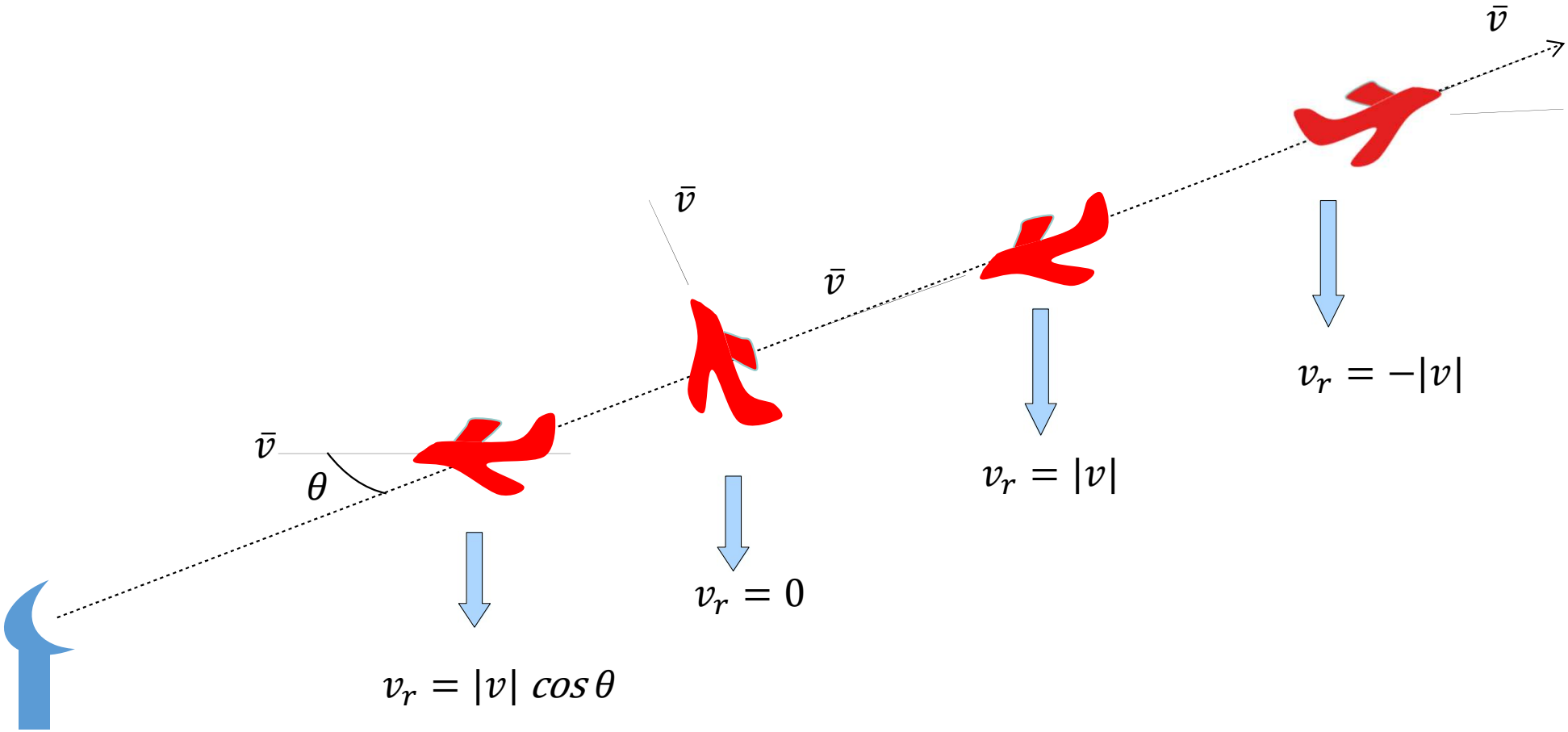
where $f_d = 2\frac{v_r}{\lambda} = 2f\frac{v_r}{c}$ *(Doppler frequency)*

and transmitting frequency: f_t

Re−radiated (echo) signal frequency: $f_r = f_t + f_d$



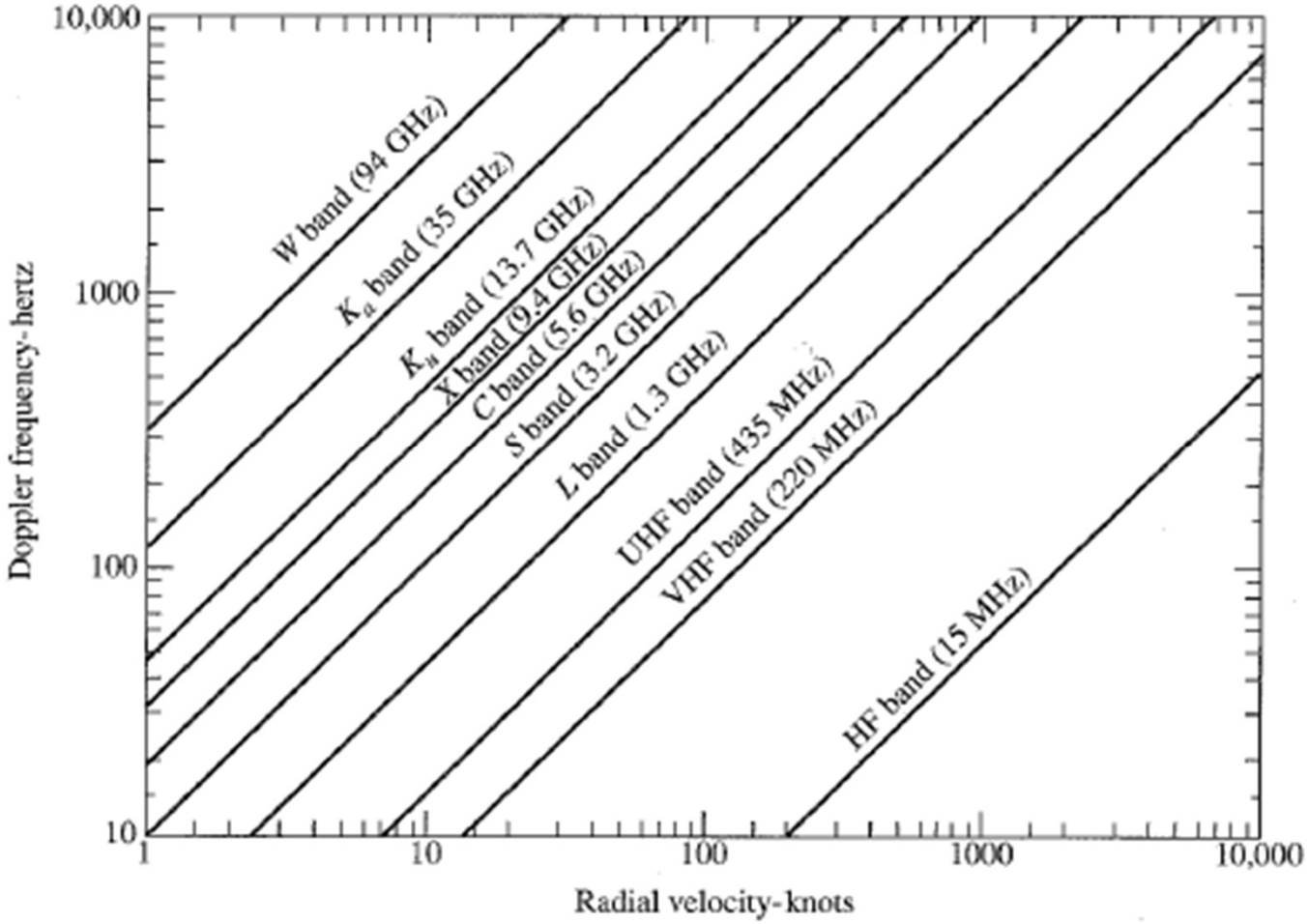
Direction of motion in Doppler shift



Doppler frequency [Skolnik]

- 1 knot
- ≅0.5 m/s
- ≅1.8 km/h

$$f_d = 2\frac{v_r}{\lambda}$$

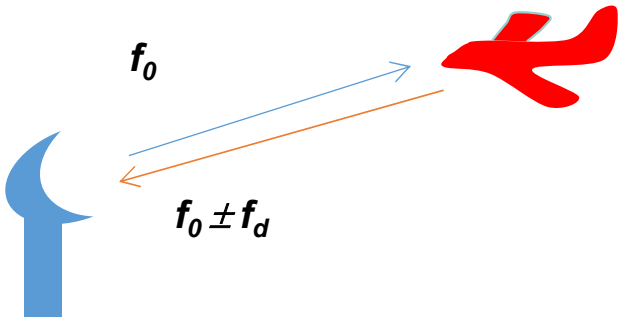


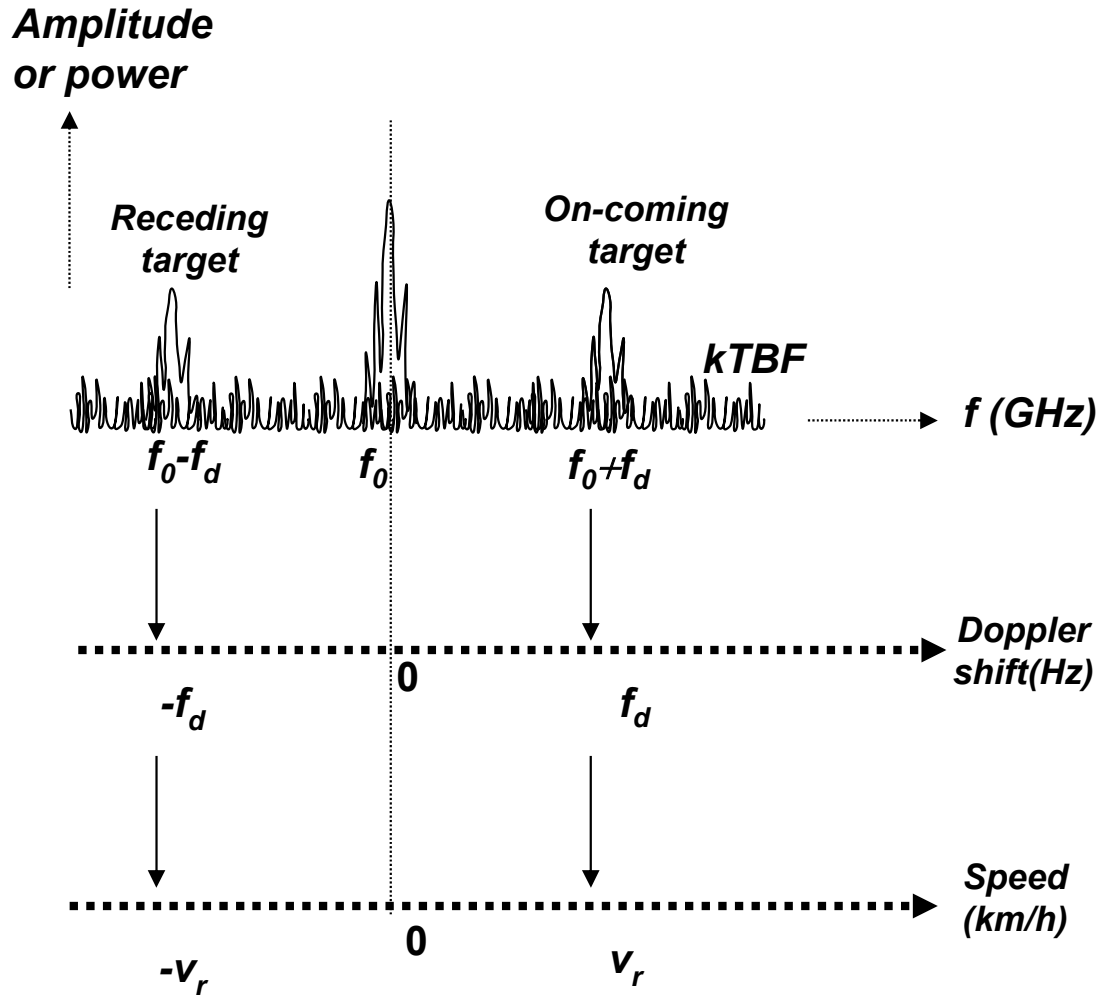
[redacted]le

[redacted] the Doppler frequency measured by the radar in the figure (f=9GHz).



Doppler shift/velocity space





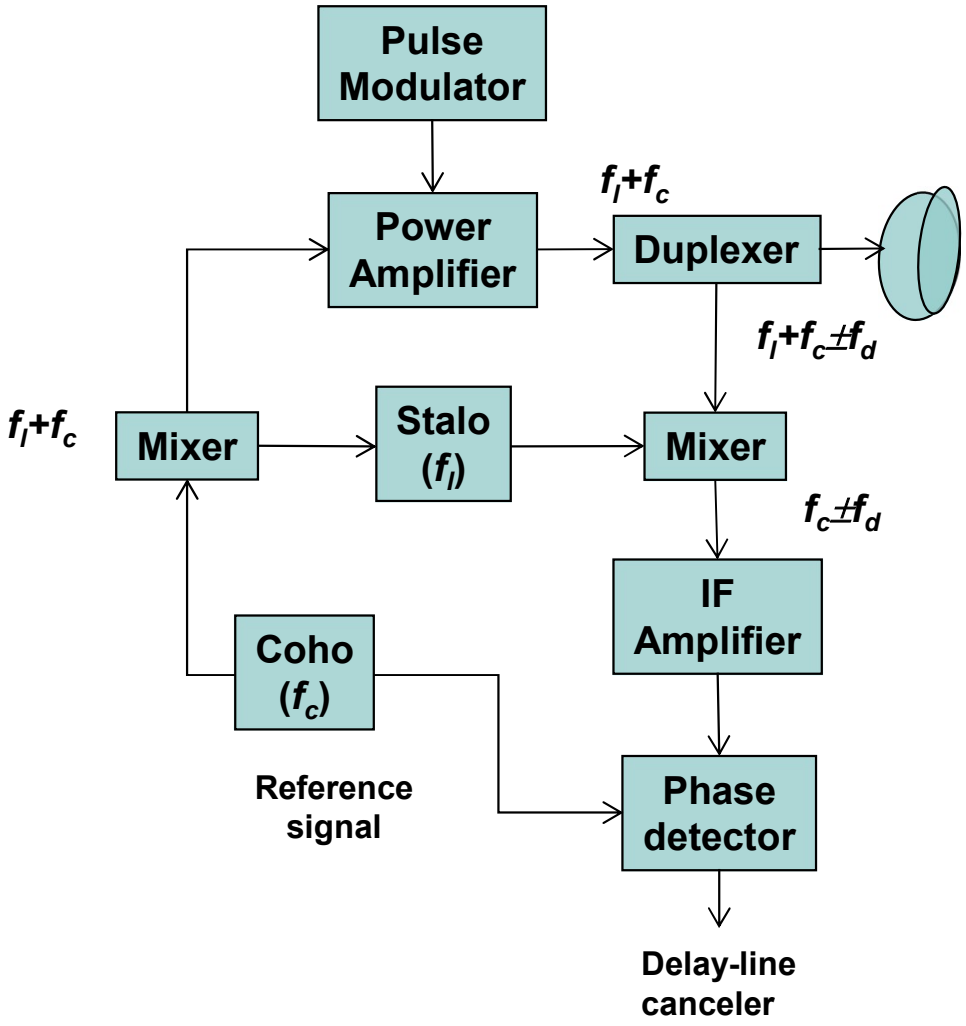
Traffic radars: X, Ku, K and Ka bands

Comparison: MTI, Pulse Doppler and CW

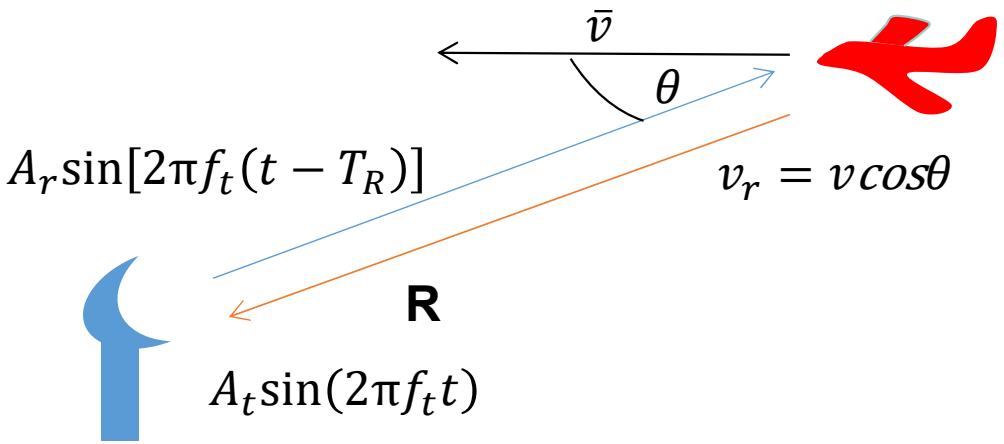
- MTI (moving target indication) radar
  - •Low PRF (high PRI)
  - •No range ambiguity
  - •Doppler(speed) ambiguity ??(stationary clutter elimination)
- Pulse Doppler radar
  - •High PRF (low PRI)
  - •Range ambiguity
  - •No Doppler(speed) ambiguity ??
- MediumPRF
  - •Compromising solution for range/doppler ambiguities
- CW Doppler:
  - •Measurement of speed in clutter (no range measurement)
  - •Synthetic aperture radar(SAR) or Inverse SAR (ISAR)
  - •Meteorological radars

MTI operation

- Need more stable local OSC, no phase change between the pulses: STALO(stable local oscillator)
- Use phase detector instead of amplitude detector after IF stage (a mixer type device)
- COHO (coherent oscillator) generates reference signal having the phase of the transmitter signal
- Use delay line canceler for detecting moving targets
- Delay line canceler subtracts one sweep from the previous one



Extracting f_d : Pulse Doppler Rx processing



Heterodyning(mixing) the received signal with $A_{ref}\,sin(2\pi f_t t)$, *the difference signal* is

$$V_d = A_d cos\left(2\pi f_d t - \frac{4\pi R_0}{\lambda}\right)$$

For stationary targets: $f_d = 0$

the output signal → constant (no motion detected)

Round−trip *time* :$T_R = \dfrac{2R}{c}$

changing range(*at constant speed*):$R = R_0 - v_r t$

$$V_r = A_r sin\left[2\pi f_t\left(1 + \frac{2v_r}{c}\right)t - \frac{4\pi f_t R_0}{c}\right]$$

The received signal frequency changes with a factor :

$$\frac{2f_t v_r}{c} = \frac{2v_r}{\lambda} = f_d$$

f_d*: doppler frequency shift*

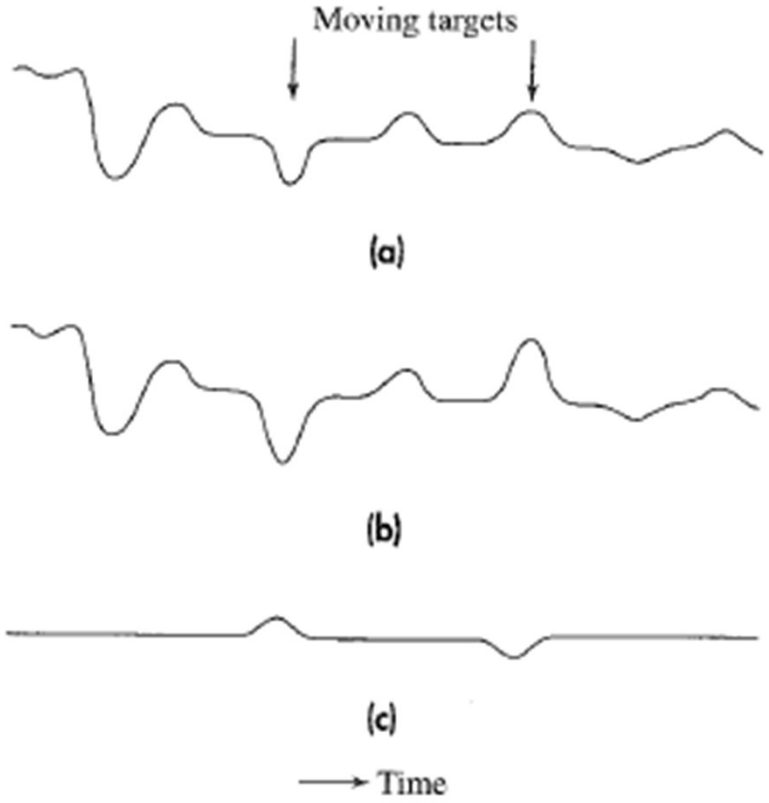
Long pulse width

+

Large doppler shift

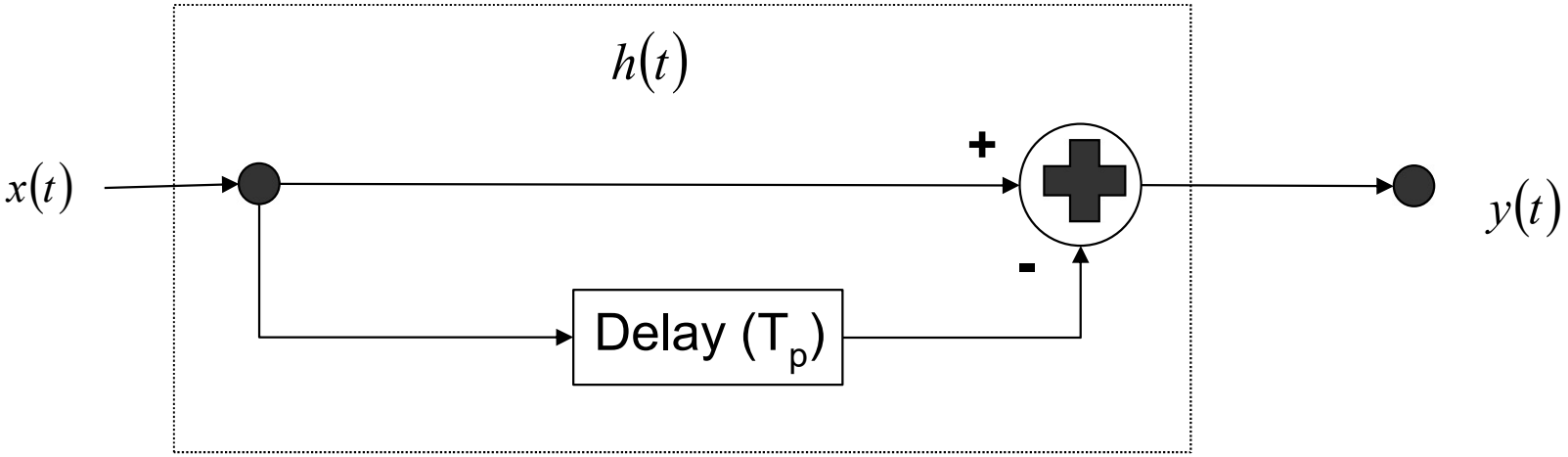
Possible to detect doppler frequency shift on the basis of frequency change within a single pulse !

MTI : DLC subtracting



- Moving target echoes with a doppler frequency shift
- Others have fixed frequency(no doppler shift)
- Subtracting one sweep of the echo signal from the previous one gives zero for fixed targets echoes, while it gives moving target echoes
- Mostly, this process performed digitally

DLC:Derivation of frequency response



Impulse Response

$$h(t) = \delta(t) - \delta(t - T_p)$$

Transfer function

$$H(f) = 1 - \exp(-j2\pi f_d T_p)$$

$$|H(f)| = 2\sin(\pi f_d T_p)$$

MTI:DLC response (time domain filtering)

detector *output for pulse 1:*

$$V_1 = k\sin(2\pi f_d t - \phi)$$

detector output for pulse 2 (with PRI delay):

$$V_2 = k\sin\left[2\pi f_d\left(t - T_p\right) - \phi\right]$$

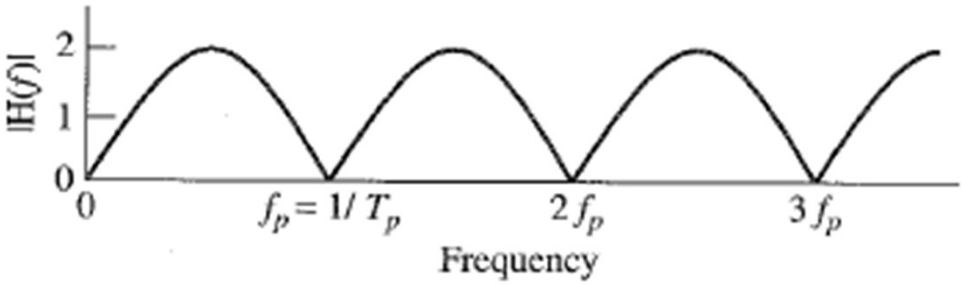
output of delay line canceler (subtraction: $\sin(a) - \sin(b)*$*)*

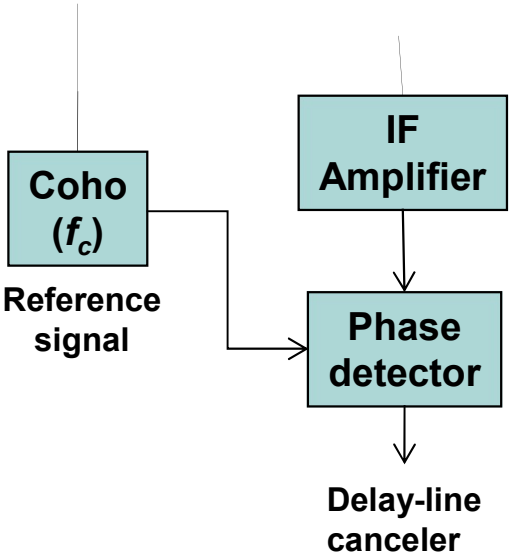
$$V_1 - V_2 = 2k\sin\left(\pi f_d T_p\right)\cos\left[2\pi f_d\left(t - \frac{T_p}{2}\right) - \phi\right]$$

$$Transfer\ function$$

$$H(f) = 2\sin\left(\pi f_d T_p\right)$$

$$|H(f)| = |2\sin\left(\pi f_d T_p\right)|$$





1) zero response at multiple of f_p
2) slowly moving clutter to be fall into passband of the filter (no delta function at f=0)

$*\sin(a) - \sin(b) = 2\sin[(a-b)/2]\cos[(a+b)/2]$

MTI: Blind Speeds (doppler ambiguity)

$$|H(f)| = |2\sin(\pi f_d T_p)|$$

Delay line canceler: a simple doppler filter

$$H(f) = 0 \ \textit{whenever} \ \pi f_d T_p = m\pi \ \ \textit{where} \quad m = 0, \pm 1, \pm 2, \pm 3, ..$$

giving

$$f_d = \frac{2v_r}{\lambda} = \frac{n}{T_p} = nf_p(\pi f_d T_p)$$

$$\textit{where } n = 0,1,2,3,.., \quad \textit{since } \ f_d, T_p > 0$$

Radial velocity producing zero response in Delay line canceler: Blind speed

$$v_r^n = \frac{n\lambda}{2T_p} = \frac{n\lambda f_p}{2}$$

$$\textit{where } v_r^n \textit{ is the} n - \text{th } \textit{blind speed } (n = 1,2,3,\dots)$$

First blind speed is important, others are multiple of the first one !

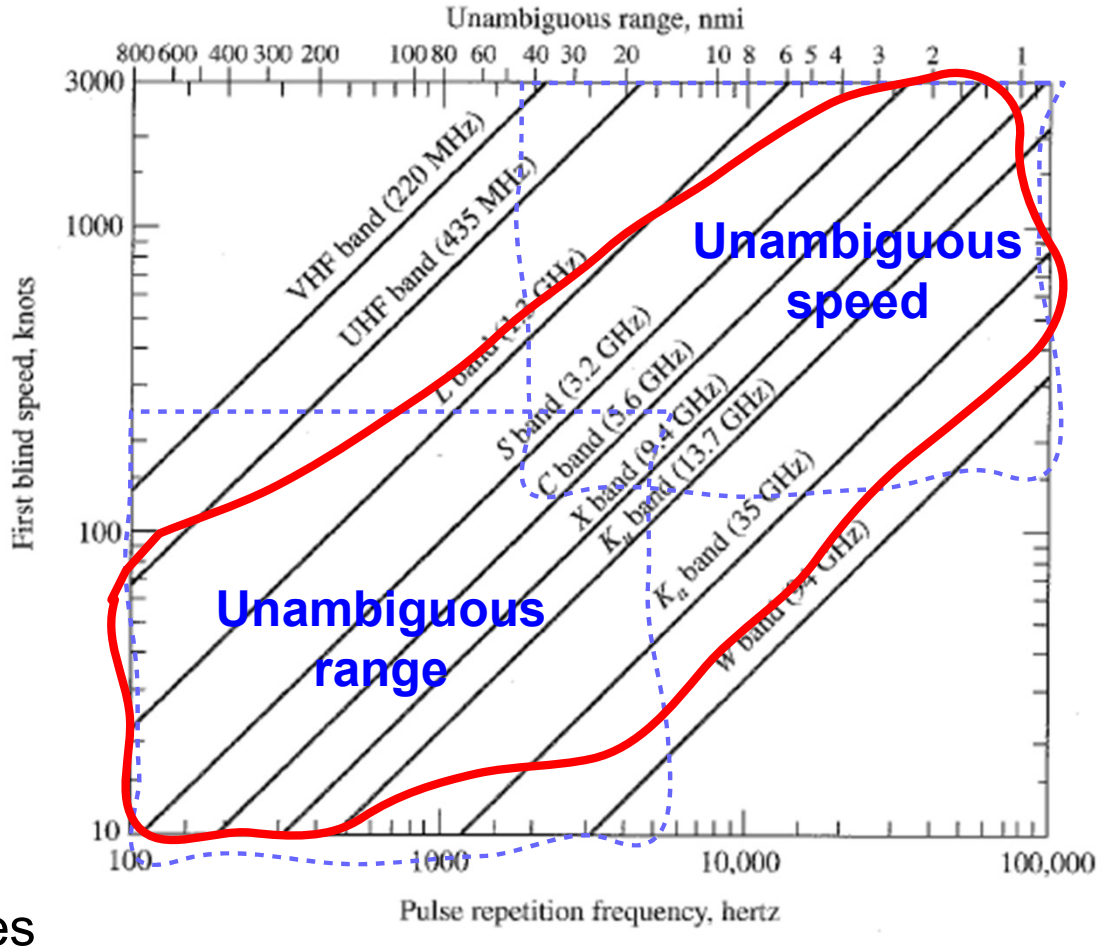
Blind speeds

$$v_r = \frac{\lambda f_p}{2}$$

To reduce the effects of blind speeds:

- Low frequency operation
- High PRF
- Multiple PRF
- Multiple carrier (RF)
- Or combinations of two or more

Choosing lower frequencies
-low resolution
-crowded spectrum
-bad coverage at low altitudes



Blind speed, unambiguous range and PRF

PRI types

Importance of PRI

- one of the distinguishing parameter of a radar
- allows radar detection systems (RWR, ESM) to identify type of the radar (so connected weapon system!)

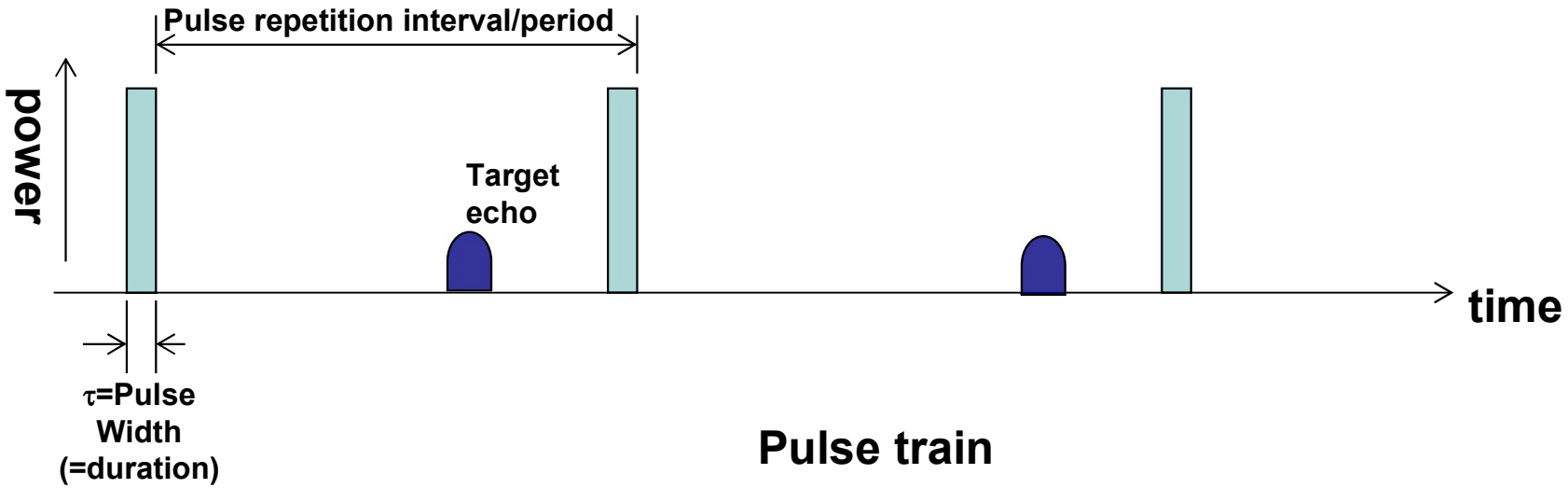
Typical PRI types

- Stable (constant)
- Staggered
- Dwell and switch
- Linear sliding
- Sinusoidal
- Jittered
- Scheduled(multi function operation)

Radars might also use Pulse Groups Repetition Intervals (PGRI), to increase range(pulse width) and doppler resolution(duration of pulse group) together

EWIRDB

Radar signal: pulse train

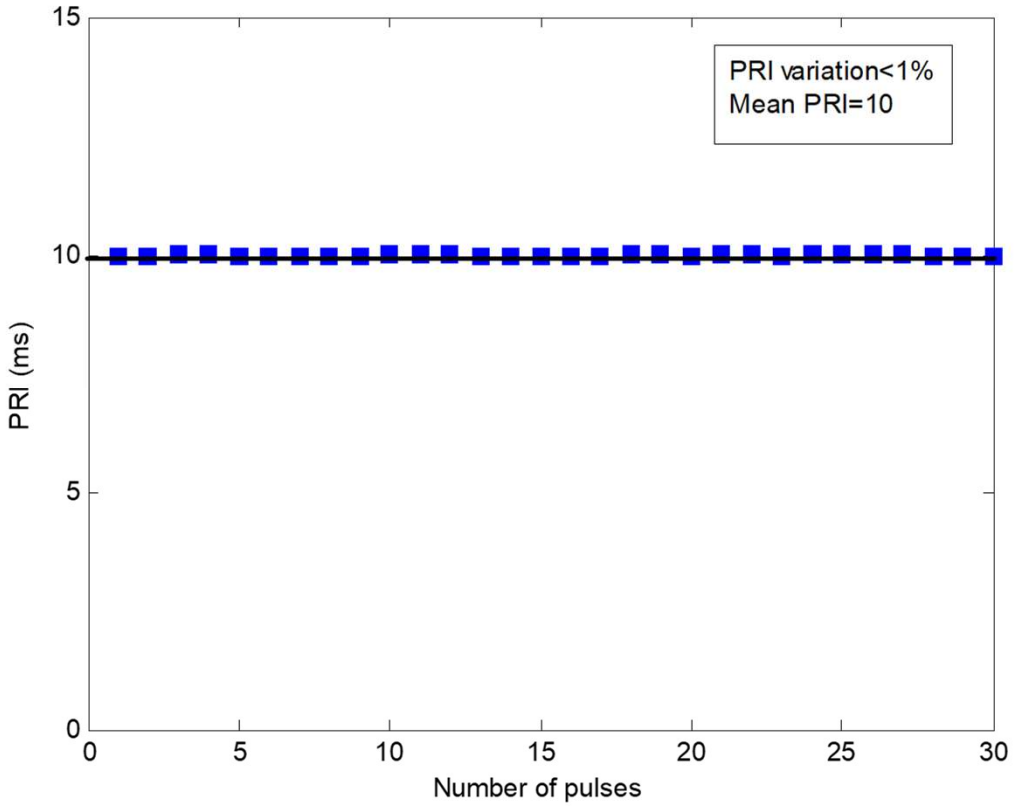


Pulse train

- $T_P = 1/f_P$ =Pulse repetition interval (PRI)
- f_P=pulse repetition frequency(PRF)
- P_{av}=average power=$P_t \tau/T_P = P_t \tau f_P$
- τ/T_P =Duty cycle

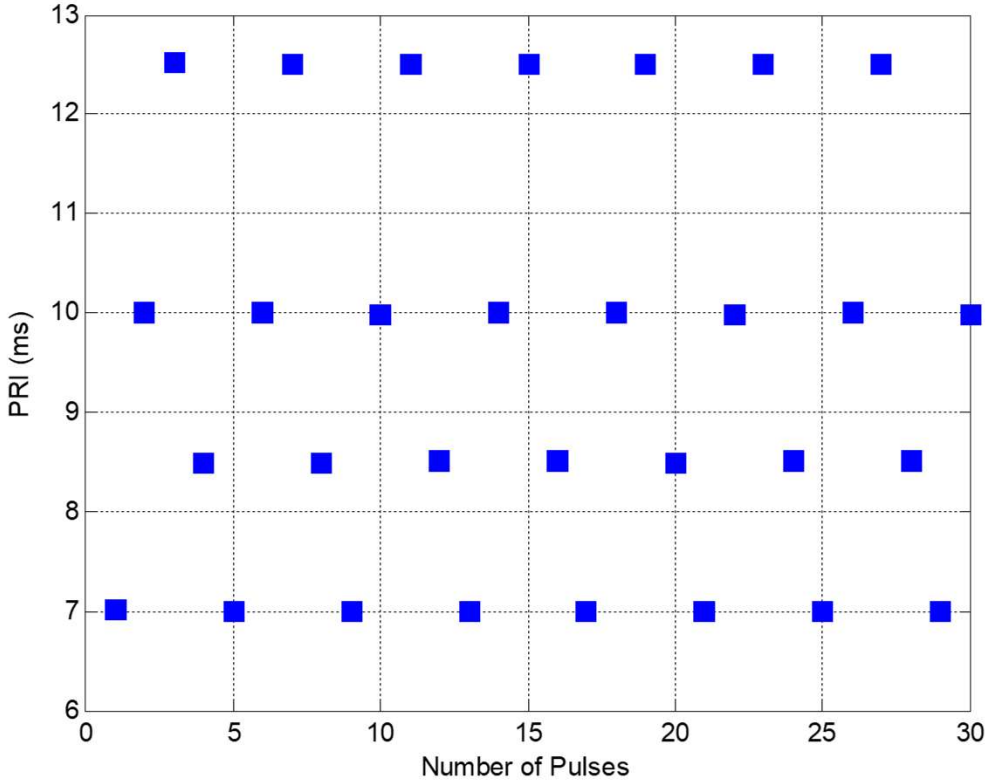
Stable (or constant) PRI

- Common search or tracking radars
- Variation (unintentional=jitter) is usually less than 1%
- Very stable with MTI doppler systems



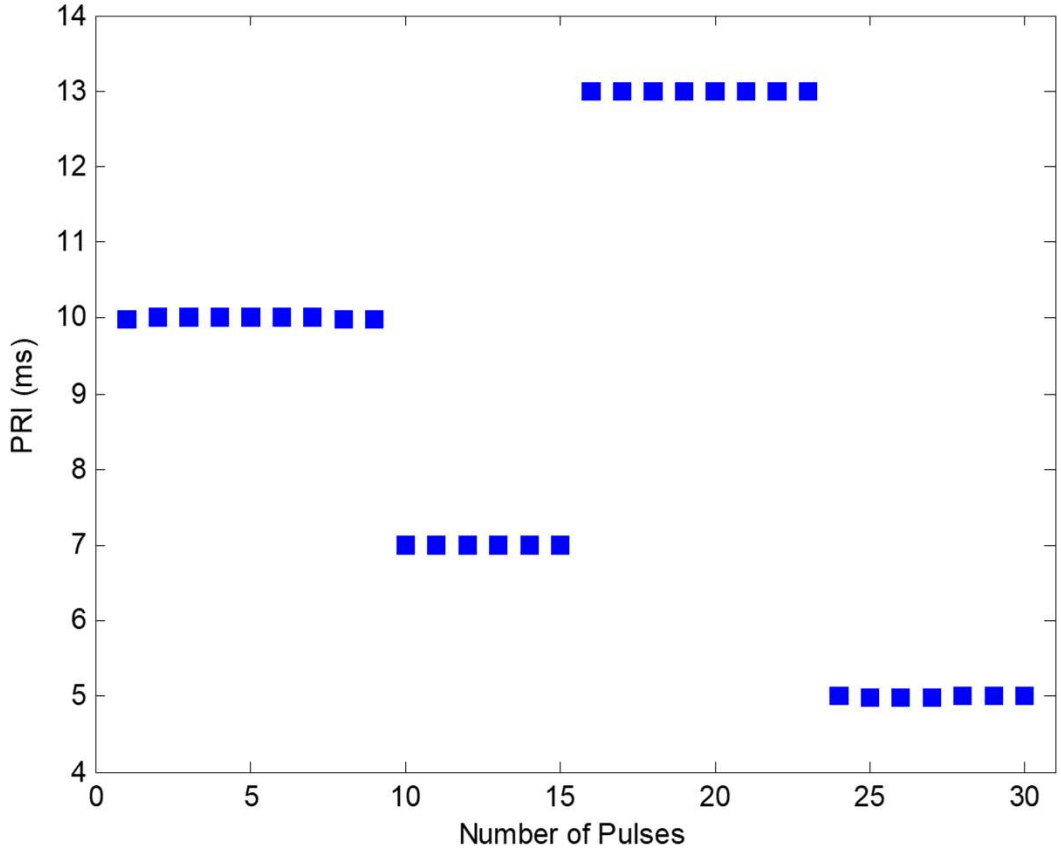
Staggered (see next slides)

- Several stable PRIs switched on a pulse to pulse basis in a periodic pattern
- Eliminate blind speeds in MTI systems



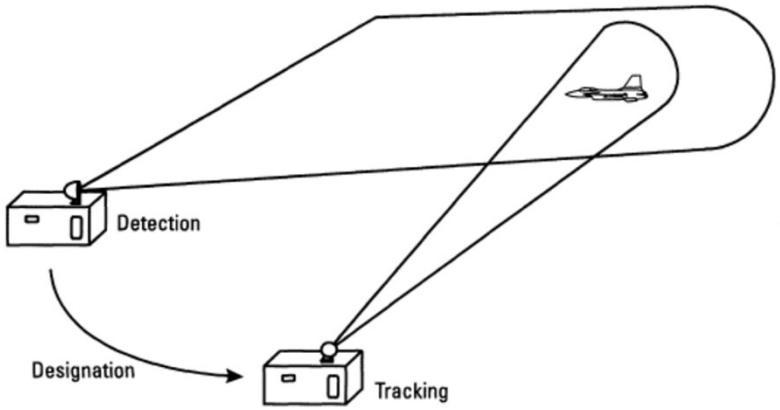
Dwell and switch

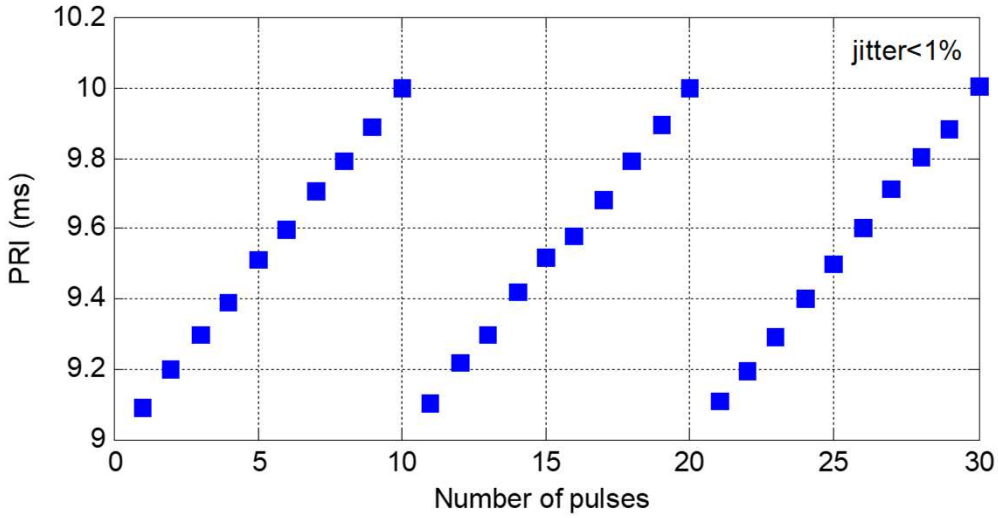
- Bursts of pulses with several stable PRIs switched from one burst to the next
- Resolve velocity (and range-eclipsing) ambiguities in pulse Doppler radar

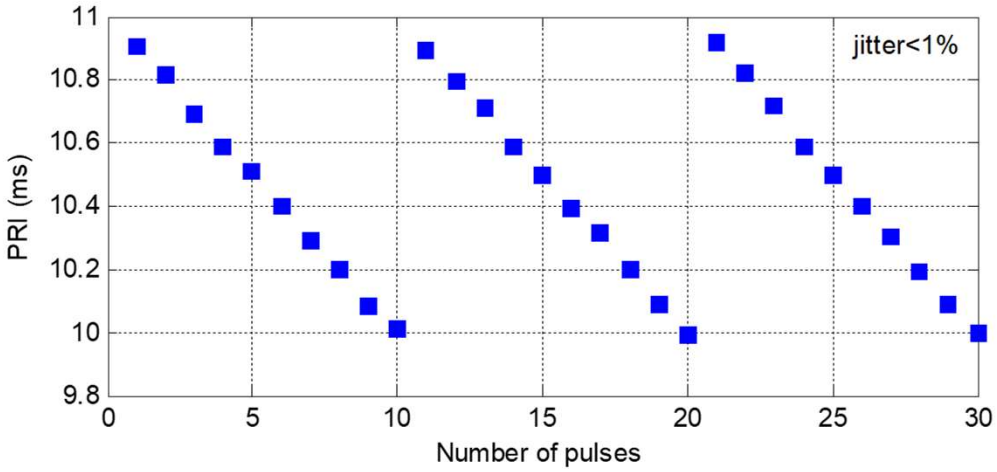


Linear sliding

- Constant altitude coverage during elevation scanning
- Maximum and minimum PRI determines the unambiguous range at different elevation angles (6 times difference)

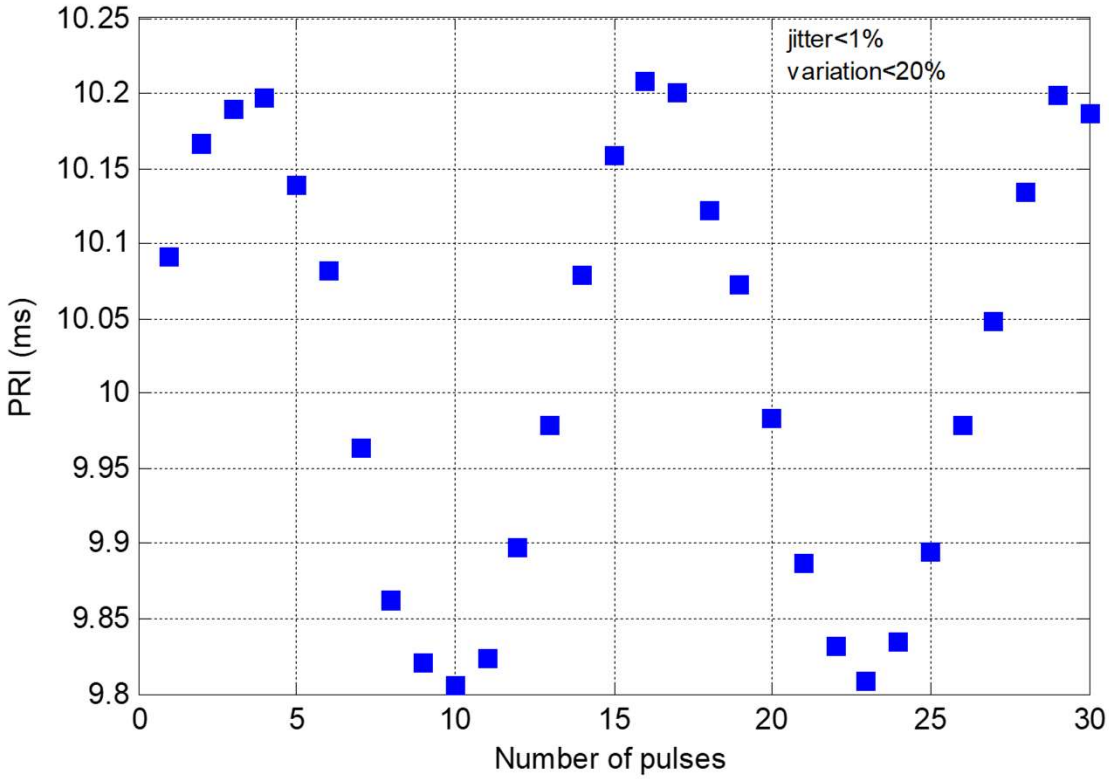






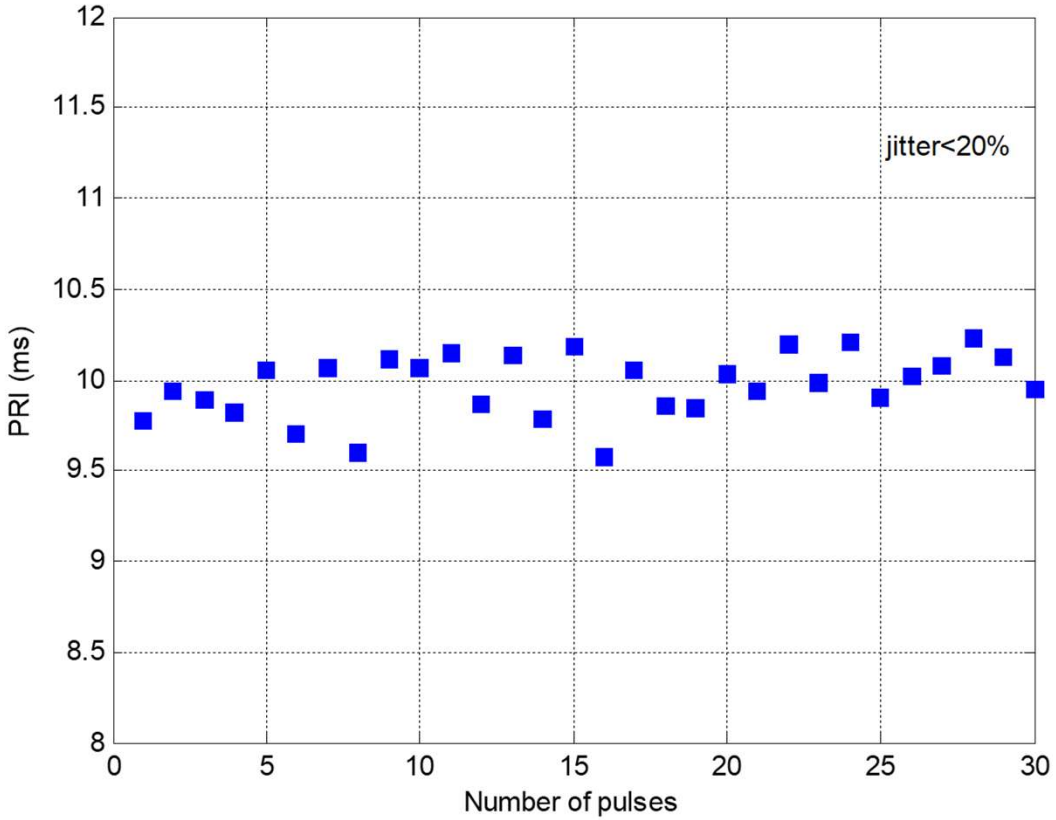
Sinusoidal (wobulated)

- Missile quidance, ranging
- Conscan tracking
  - ✓ no variation when the missile headed toward the target
  - ✓ Missile needs a reference signal
  - ✓ PRI variation might be used as a reference
- Up to 5% variation in practice (50Hz or more)



Jittered

- Intentional random variation (up to 30%)
- Discrete intervals
- Intervals in specified range
- Against some jamming signals



Staggered PRF: reducing blind speeds

Very *popular in air traffic radar*

$f_1 = first\,PRF(f_{p1})$

$f_2 = second\,PRF(f_{p2})$

First blind speed: $2f_1 = 3f_3$

Blind speed due to f_1 *is compansated by* f_2

first blind speed is doubled $(f_1 \rightarrow 2f_1)$

prf *changed from*

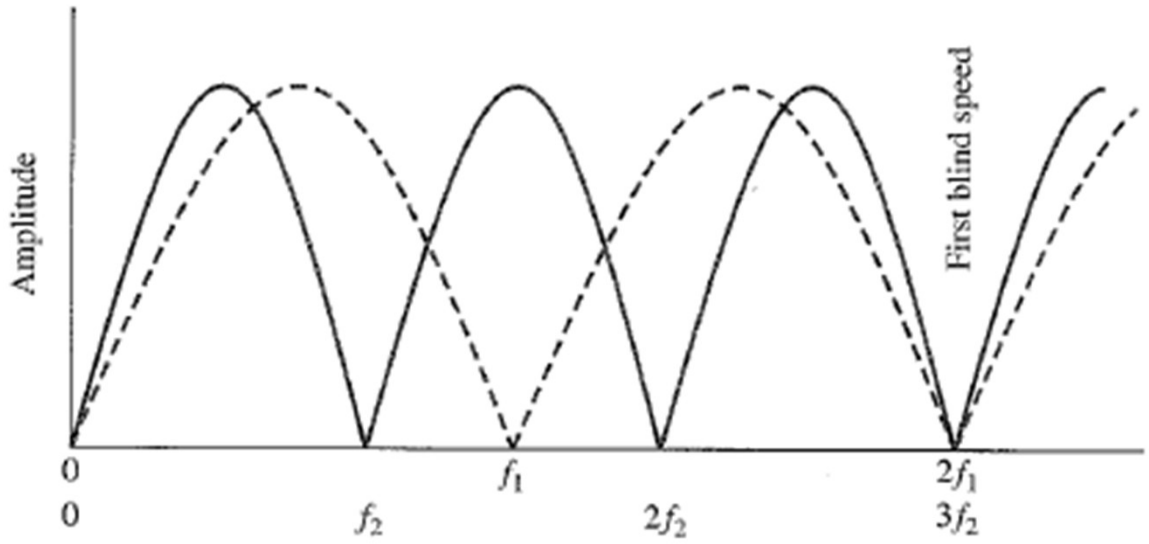
a) scan to scan

b) dwell to dwell

c) pulse to pulse

dwell: time on target

(scanning time of a beamwidth)



Frequency response
of a single DLC for two different PRFs

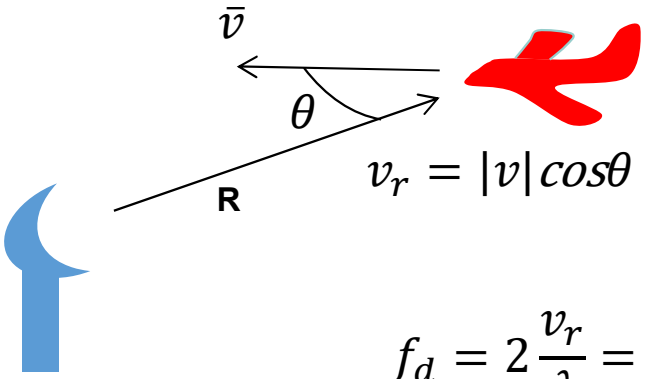
Pulse Doppler Radar: AWACS

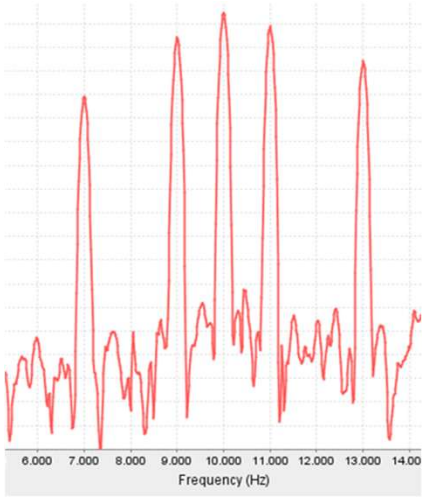
- all-weather, surveillance, Command and Control and communications (C^3)capability
- Doppler velocity sensor
- An S-band (2 to 4 GHz) overland surveillance radar
- Very Low Frequency (VLF - 3 to 30 kHz) hyperbolic radio navigation system
- Identification Friend-or-Foe (IFF) interrogator
- HF (3-30 MHz), three AM and one FM based VHF (30-300 MHz) and seven UHF (0.3-3GHz) band radios
- Computers, Situation Display Consoles
- dedicated workstations for a duty officer, a radar maintenance technician, a computer operator and a communications specialist
- Radar coverage: 312.000 km^2 (at 9.150m)
- Radar range: 400 km to 520 km



Peace eagle project

Doppler effects (fixed radar) [Skolnik]

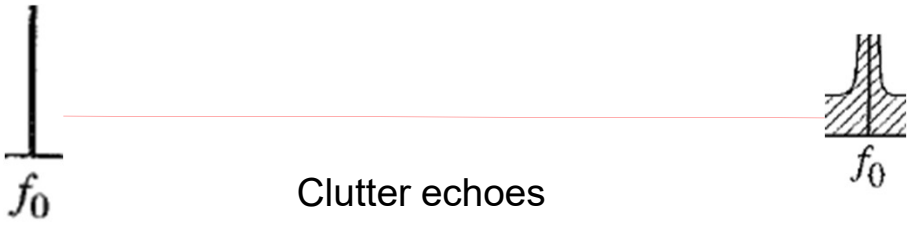




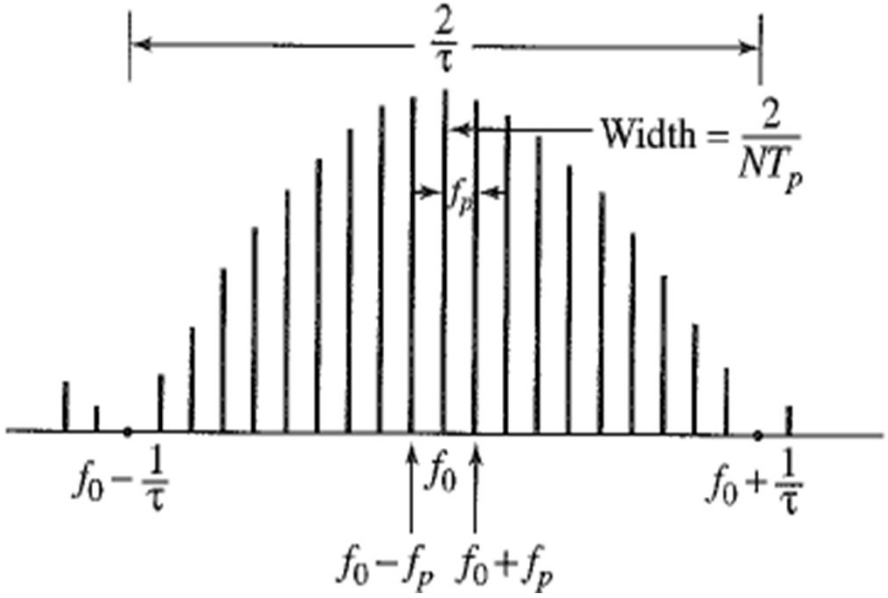
$$f_d = 2\frac{v_r}{\lambda} = 2f\frac{v_r}{c}$$

$A_t \sin(2\pi f_t t)$

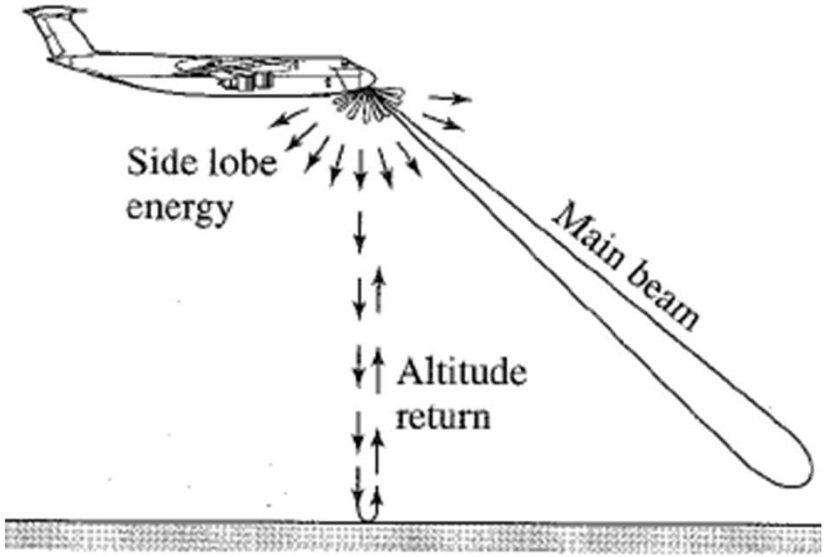
$$A_r sin\left[2\pi f_t\left(1+\frac{2v_r}{c}\right)t\right]$$

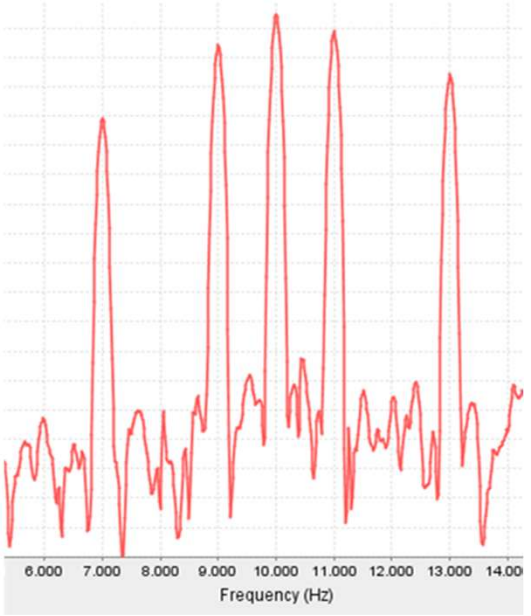


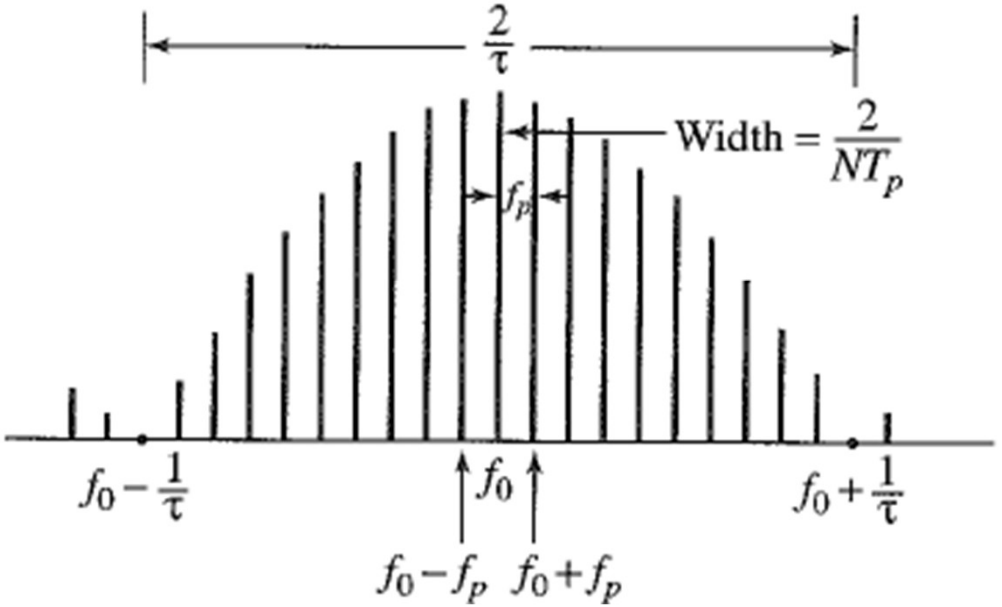
Clutter echoes



Doppler effect: moving radar [Skolnik]

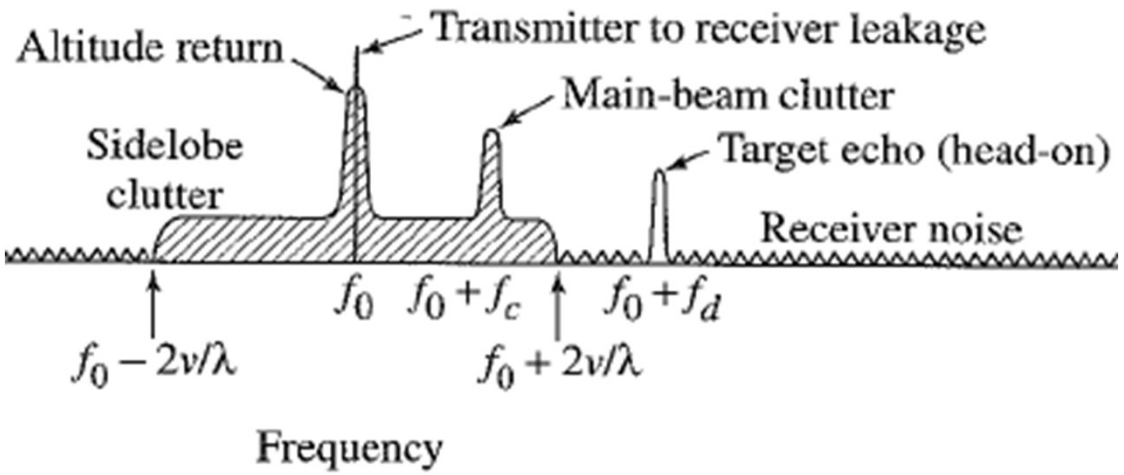


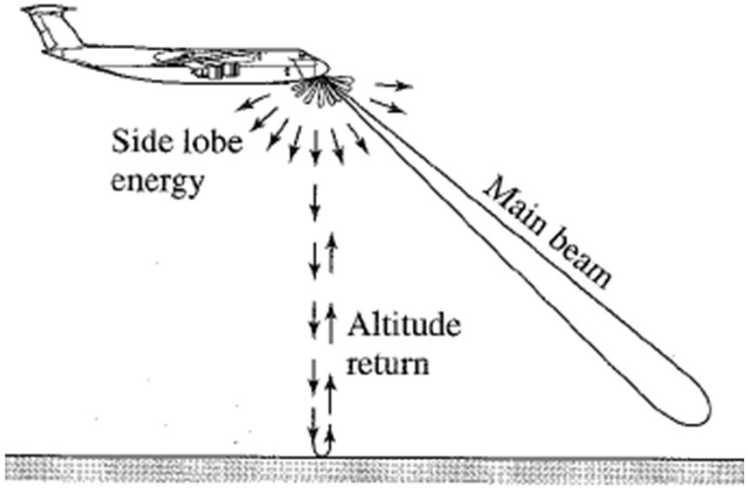




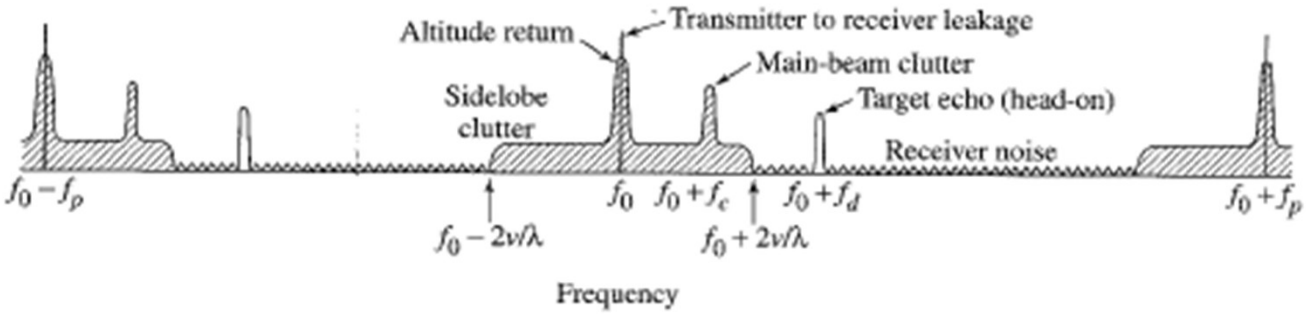
Doppler echoes: spectrum [Skolnik]

v: *radar's absolute velocity*

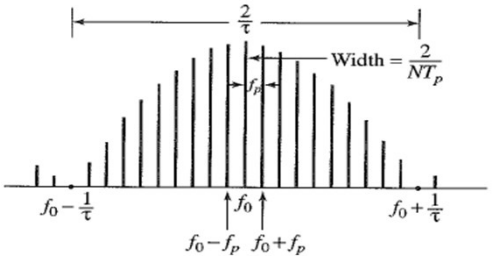




Main beam scans in angle, main beam clutter moves over sidelobe clutter



High prf: 100-300 kHz
-Low duty cycle (0.3-0.5)
-One range gate
-No doppler ambigutity



Radar requirements [Skolnik, Radar Handbook]

Radar Application	Requirements
Airborne or spaceborne surveillance	Long detection range; accurate range data
Airborne interceptor or fire control	Medium detection range; accurate range, velocity, and angle data
Ground-based surveillance	Medium detection range; accurate range data
Battlefield surveillance (slow-moving target detection)	Medium detection range; accurate range, velocity data
Missile seeker	Short detection range; accurate velocity and angle rate data; may not need true range information
Surface-based weapon control	Short range; accurate range, velocity data
Meteorological	Good velocity resolution
Missile warning	Short detection range; very low false-alarm rate